



NETWORK INTRUSION DETECTION · CIC-IDS2017 · BINARY CLASSIFICATION

Network Intrusion Detection Using Machine Learning and Deep Learning

Distinguishing Normal Traffic from Attack Traffic in Network-Flow Data

Under the supervision of:

DR. Heidi Hassan & Dr. Ahmed Farouk

Researchers at Electronics Research Institute (ERI)

Prepared by:

Rahma Abdelaziz Omar

Communication Systems Engineering Student
Benha National University



Why Network Intrusion Detection Matters



Traffic at scale

Modern networks generate massive volumes of flow data every second — far beyond what a human operator can inspect.



Manual review doesn't scale

Manually distinguishing malicious flows from normal traffic is slow, error-prone, and cannot keep pace with live networks.



Automated detection is required

Intrusion Detection Systems (IDS) continuously monitor traffic and flag suspicious flows in real time.



ML/DL learns the pattern

Machine Learning and Deep Learning models learn statistical patterns of attacks directly from flow features, instead of relying on fixed signatures.

DETECTION PIPELINE

Network
Traffic



Feature
Extraction



ML / DL
Model



Normal /
Attack

Project Objectives



01

Analyze Network Traffic

Explore the CIC-IDS2017 flow-level dataset and understand its structure and class balance.



02

Preprocess & Clean

Check for missing/infinite values, corrupted rows, and train/test leakage before modeling.



03

Build Detection Models

Train binary classifiers to separate Normal Traffic from Attack Traffic.



04

Compare ML vs DL

Benchmark traditional Machine Learning against a Deep Learning (MLP) approach.



05

Identify Best Model

Select the strongest model using multiple evaluation metrics, not accuracy alone.

The Dataset — CIC-IDS2017

2,520,751

Total flow records

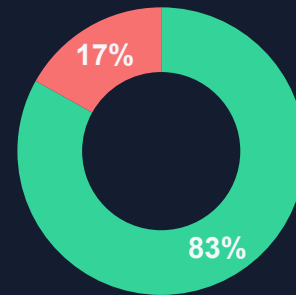
52

Network-flow features

83.1% / 16.9%

Normal vs Attack split

CLASS DISTRIBUTION



■ Normal (0) ■ Attack (1)

REPRESENTATIVE FEATURE GROUPS

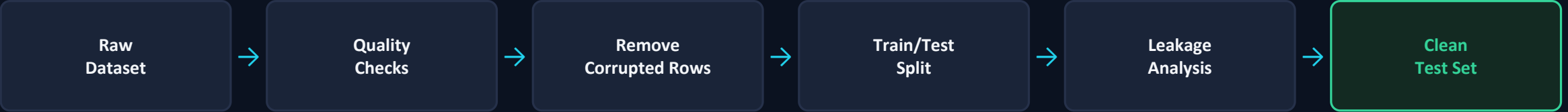
- Packet statistics (length, size)
- Flow duration
- Inter-arrival times (IAT)
- TCP flag counts
- Packet length statistics
- Destination port



Target label: 0 = Normal Traffic 1 = Attack Traffic

Binary classification was chosen because several individual attack classes in CIC-IDS2017 are extremely rare — grouping them into a single "Attack" class gives a more reliable initial evaluation.

Data Preprocessing & Data Quality



Missing / Infinite Values

0 missing values and 0 infinite values across the full dataset — no imputation required.

Corrupted Rows Removed

35 rows with impossible negative header-length / segment-size values were removed before splitting (2,520,751 → 2,520,716).

Class Balancing

Only the training set was undersampled to a 1:1 ratio (340,555 / 340,555). The test set kept its natural, real-world class distribution.

Train/Test Feature-Level Leakage Check

156 rows in the test set had feature values that exactly duplicated a row already seen in training (label-independent match) — these were removed from the test set only.

504,144

Original test size

156

Overlapping rows removed

503,988

Final clean test size

Identical feature rows appearing in both train and test can cause data leakage and overly optimistic evaluation — this is a feature-level duplicate check, distinct from exact duplicates that also match on label.

Models Evaluated

TRADITIONAL MACHINE LEARNING



Decision Tree

Baseline, interpretable tree-based model that splits on feature thresholds.



Random Forest

Ensemble of many decision trees — a strong, robust baseline for tabular data.



XGBoost

Gradient-boosted trees — a leading benchmark for structured/tabular classification.



MLP Neural Network

Used to directly compare Deep Learning against traditional ML on the same tabular flow data.

```
Dense(128, ReLU)
Dropout(0.3)
Dense(64, ReLU)
Dropout(0.3)
Dense(32, ReLU)
Dense(1, Sigmoid)
```

Evaluation Metrics

Accuracy

Share of all predictions that are correct.

Precision

Of predicted attacks, how many were real attacks.

Recall

Of real attacks, how many were correctly caught.

F1-Score

Harmonic mean balancing Precision and Recall.

ROC-AUC

Discrimination ability across all thresholds.

PR-AUC

Precision/Recall trade-off — most informative for the rare, positive Attack class.



Why Accuracy Alone Is Not Enough

A False Negative means an **actual Attack is classified as Normal** — the most dangerous type of error for an intrusion detection system, since the threat passes through undetected.

Recall is prioritized to minimize missed attacks. **F1-Score** balances Precision and Recall. **PR-AUC** is especially informative here since attacks are the minority class in the natural test distribution.

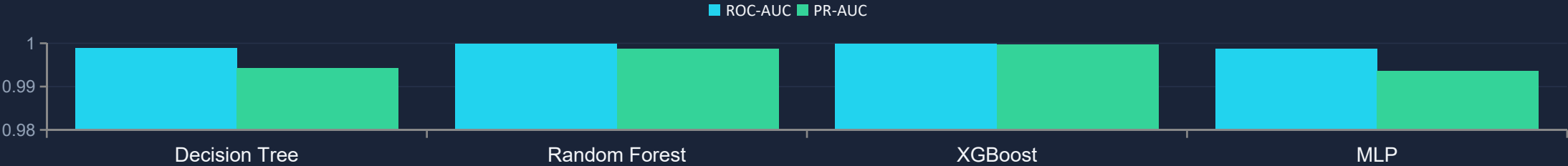
Final Model Results — Clean Test Set

Evaluated on the clean test set (n = 503,988), after removing feature-level train/test overlap.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Decision Tree	1.00	1.00	1.00	1.00	0.9990	0.9943
Random Forest	1.00	1.00	1.00	1.00	0.9999	0.9988
XGBoost	1.00	0.99	1.00	1.00	0.99997	0.9998
MLP Neural Network	0.98	0.88	0.99	0.93	0.9987	0.9936

Note: Accuracy/F1 values shown as 1.00 are rounded to two decimals — they are not literally perfect. The confusion matrices (next slide) show the models still make a small number of errors.

ROC-AUC & PR-AUC BY MODEL



Confusion Matrix — Best Model (XGBoost)

	Predicted Normal	Predicted Attack
Actual Normal	418,299 True Negative	575 False Positive
Actual Attack	46 False Negative	85,068 True Positive



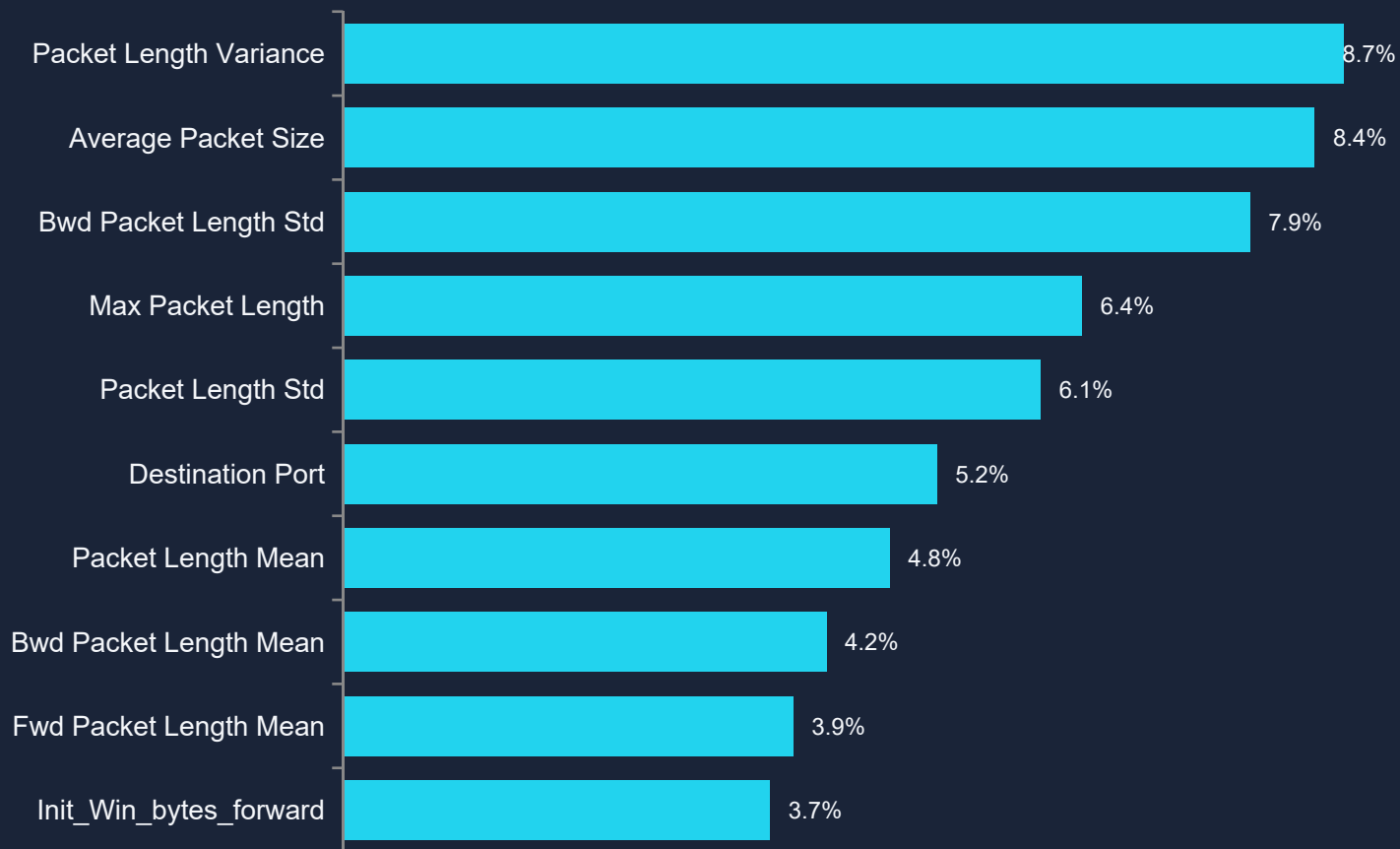
Only 46 attack samples were classified as normal.

Out of 85,114 real attack flows in the clean test set, XGBoost missed only 46 — a False Negative rate of about 0.05%.

TN 418,299 **FP** 575

FN 46 **TP** 85,068

Feature Importance — Random Forest



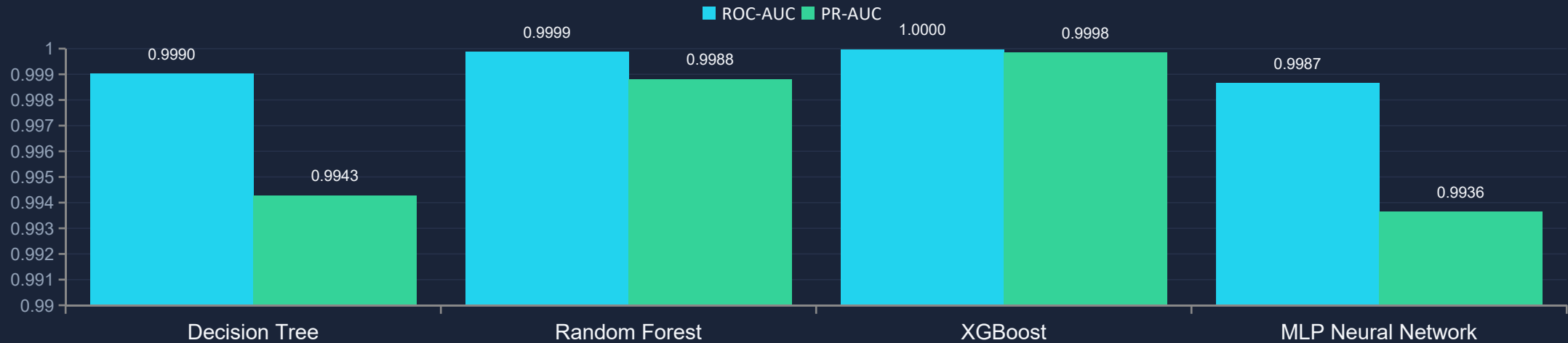
What This Tells Us

Packet-size and length-distribution statistics dominate the top features — attack flows tend to have distinctive packet-size patterns compared to normal traffic.

Important: feature importance shows predictive contribution, not causation.

Model Comparison — Discrimination Ability

Full ROC and Precision-Recall curves for all four models are available in the notebook appendix. Summarized here by final AUC scores on the clean test set.



XGBoost leads

Highest ROC-AUC (0.99997) and PR-AUC (0.9998) among all four models.

MLP: strong recall, more false alarms

Competitive attack recall, but lower precision and F1 than the tree-based models.

Tree ensembles suit tabular data

Random Forest and XGBoost both handle structured, flow-level network features particularly well.

Conclusion & Future Work

CONCLUSION

- ML and DL models can effectively separate attack traffic from normal traffic on CIC-IDS2017.
- Tree-based ensemble methods achieved the strongest overall results on this tabular flow data.
- XGBoost delivered the best overall discrimination — highest ROC-AUC and PR-AUC.
- Random Forest closely followed, with excellent precision and recall.
- MLP reached strong attack recall but produced more false positives than the tree models.

FUTURE WORK



Multiclass attack-type classification



Hyperparameter optimization



Feature selection



More advanced neural network architectures



Real-time intrusion detection



Evaluation on additional datasets



Best Overall Model: **XGBoost** — ROC-AUC 0.99997 · PR-AUC 0.9998 · only 46 missed attacks out of 85,114



Thank You

Questions & Discussion

Rahma Abdelaziz Omar · Network Intrusion Detection Using ML and DL · CIC-IDS2017

